Document Scanner

Automatic Perspective Correction Using Classical Computer Vision

Hari Haran Manda
MSML640 - Computer Vision
Spring 2026

The Problem

When you photograph documents with your smartphone:

- Document appears tilted or at an angle
- Shadows and lighting make it hard to read
- Background clutter is visible
- Not professional or scanner quality

The Solution: Automatic Document Scanner

Automatically detect document boundaries and apply perspective correction - like a mobile scanner app, but built from scratch using classical computer vision!

✓ Works in under 1 second
✓ No expensive equipment needed
✓ Built with Python & OpenCV
✓ Professional results

How It Works - 5 Steps

1. **Grayscale Conversion**
   Simplify image to focus on structure

2. **Gaussian Blur**
   Reduce noise with 5×5 kernel

3. **Canny Edge Detection**
   Find all boundaries (thresholds: 75, 200)

4. **Contour Detection**
   Identify largest quadrilateral shape

5. **Perspective Transform**
   Warp to bird's-eye view

Classical Computer Vision (Not Machine Learning) - Works immediately, no training data needed!

Results - Algorithm Success Rate

95.7%

SUCCESS

Tested on 70 Real Documents

67 scanned successfully
3 failed to detect contours

Processing Speed:

0.5 seconds per image

Document Variety:

Receipts, bills, homework, insurance cards, travel docs, manuals

Output Quality - Key Insight

Excellent Quality

~37 outputs with clear, readable text

Original photos: good lighting & contrast

Light/Washed Out

~30 outputs appear very light or faded

Original photos: over-exposed or low contrast

Critical Learning:

The algorithm can straighten perspective perfectly, but it cannot fix poor photography.

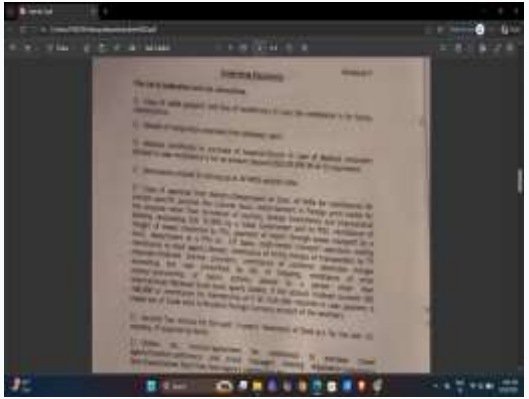
This reveals an important principle: *Data quality is just as important as algorithm quality.* Even the best CV pipeline needs well-lit, properly exposed input images to produce excellent results.

The 3 Failure Cases - Why They Failed

Hand-Held Receipt

Problem: Hand covering bottom corners

Lesson: *All 4 corners must be visible*



Browser Screenshot

Problem: No physical document boundary

Lesson: *Designed for physical docs only*

Multiple Documents

Problem: Two insurance cards in frame

Lesson: *Assumes one document per image*

What I Have Learned

✓ Algorithm Works Great When:

- Single physical document
- Clear background contrast
- All 4 corners visible
- Moderate camera angles

✗ Algorithm Struggles When:

- Hands/objects cover corners
- Digital screenshots (no edges)
- Multiple documents in frame
- Poor lighting or exposure

Key Insights:

✓ Understanding WHY algorithms fail is just as important as making them work
✓ Classical CV is powerful for well-defined, structured problems
✓ Real-world testing reveals hidden assumptions and data quality requirements

Sources & References

Technologies & Tools:

- Python 3.13 - Programming Language
- OpenCV 4.8+ - Computer Vision Library
- NumPy - Numerical Computing Library

Algorithms Referenced:

- Canny Edge Detection (J. Canny, 1986)
- Perspective Transformation (Homography)
- Contour Detection & Approximation

📂 **Full Code & Dataset: github.com/Nani3434/Document-Scanner-MSML640**

Thank You!